

1) Write a program to Calculate the percentage of student based on marks of any 5 Subjects.

- Start

- Input the marks of the 5 Subjects

Sub 1 = 80

Sub 2 = 65

Sub 3 = 72

Sub 4 = 85

Sub 5 = 90

- Input maximum marks

Maximum Marks = ~~300~~ 500

- Calculate the total marks by adding all 5 subjects

$$\begin{aligned}\text{Total marks} &= \text{sub 1} + \text{sub 2} + \text{sub 3} + \text{sub 4} + \text{sub 5} \\ &= 80 + 65 + 72 + 85 + 90 \\ &= 392\end{aligned}$$

- Perform calculation the percentage using formula

$$\begin{aligned}\text{Percentage of student} &= \left(\frac{\text{Total marks}}{\text{Maximum marks}} \right) \times 100 \\ &= \frac{392}{500} \times 100 \\ &= 78.4\end{aligned}$$

- Display the Calculated Percentage

The percentage of student is 78.4

- Stop

2) W.A.P to Calculate area of rectangle based on length and breadth.

- Start

- Input the length and breadth of rectangle
length = 25
breadth = 15

- Perform Calculation

$$\begin{aligned}\text{Area of rectangle} &= \text{length} \times \text{breadth} \\ &= 25 \times 15 \\ &= 375\end{aligned}$$

- Display the result
Area of rectangle is 375

- Stop

3) W.A.P to find quotient and remainder of two number

- start

- Input divisor & Divident

divisor = 7

divident = 18

- Perform operation

quotient = divident // divisor

= 18 // 7

= 2

remainder = divident % divisor

= 18 % 7

= 4

- Display the Result

The quotient is 2

The remainder is 4

- stop

4) W.A.P to enter P, T, R and calculate Simple Interest.

- Start

- Take input values

$$P = 8000, R = 16\%, T = 15$$

- Perform calculation using formula

$$\text{Interest} = \frac{P \times R \times T}{100}$$

$$= \frac{8000 \times 16 \times 15}{100}$$

$$= 19,200$$

- Display the result
The Simple Interest is 19,200

5) W.A.P to enter P, T, R and Calculate Compound Interest.

- START

- Input the values

$$P = 1000, T = 10, R = 12$$

- perform the calculation

$$C.I = P * \left(1 + \frac{R}{100}\right)^T - P$$

$$= 1000 * \left(1 + \frac{12}{100}\right)^{10} - 1000$$

$$= 1000 * 3.105848 - 1000$$

$$= 3105.84 - 1000$$

$$= 2105.84$$

- Display result

The compound Interest is 2105.84.

6) W.A.P to input two angles from user and find third angle of the triangle.

- START

- Input Two angles from user

angle 1 = 60

angle 2 = 90

- Calculate the third angle using the formula

$$\text{angle 3} = 180 - (\text{angle 1} + \text{angle 2})$$

$$= 180 - (60 + 90)$$

$$= 180 - 150$$

$$= 30$$

- Display the result

The third angle is 30

7) W.A.P to find the Roots of a Quadratic Equation.

- start

- Input coefficients a , b , and c

$$a = 10, b = 5, c = 6$$

- Calculate the discriminant

$$\Delta = b^2 - 4ac$$

$$= 5^2 - 4 \times 10 \times 6 = 25 - 240 = -215$$

$$= -215$$

$$= -215$$

$$= -215$$

- If check the value of the discriminant

If $\Delta > 0$: The roots are real & distinct

$$x_1 = \frac{-b + \sqrt{\Delta}}{2a}$$

$$x_2 = \frac{-b - \sqrt{\Delta}}{2a}$$

If $\Delta = 0$: The roots are real and equal

$$x_1 = x_2 = \frac{-b}{2a}$$

If $\Delta < 0$: The roots are imaginary (complex conjugates)

$$x_1 = \text{real} + i$$

$$x_2 = \text{real} - i$$

where $\text{real} = \frac{-b}{2a}$ and $\text{imaginary} = \frac{\sqrt{-\Delta}}{2a}$

Display the calculated root

$$\text{real part} = \frac{-5}{2(10)} = \frac{-5}{20} = -0.25$$

$$\text{imaginary part} = \frac{\sqrt{215}}{2(10)} = 0.73$$

Roots : $-0.25 + 0.73i$ and $-0.25 - 0.73i$

The root of quadratic equation is
 $-0.25 + 0.73i$ and $-0.25 - 0.73i$

— stop

8) W.A.P to convert days into years, weeks and days.

- start

- Input the total number of days
total = days = 1500

- Calculate years

$$\begin{aligned}\text{years} &= \text{total days} // 365 \\ &= 1500 // 365 \\ &= 4\end{aligned}$$

- Calculate remainder

$$\begin{aligned}\text{days} &= \text{total days} \% 365 \\ &= 1500 \% 365 \\ &= 40\end{aligned}$$

- Calculate weeks

$$\begin{aligned}\text{weeks} &= \text{total days} // 7 \\ &= 40 // 7 \\ &= 5\end{aligned}$$

- Calculate final days

$$\begin{aligned}\text{final days} &= \text{days} \% 7 \\ &= 40 \% 7 \\ &= 5\end{aligned}$$

- Display

The days into year are 4 and weeks is 5 and day is 5

- END

9) W.A.P. to enter base and height of a triangle and find its area.

- START

- Input the values base & height

$$\text{base} = 10$$

$$\text{height} = 30$$

- Calculate using formula

$$\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 10 \times 30$$

$$= 150$$

- Display the result
Area of triangle is 150

- STOP

10) W.A.P to calculate area of an equilateral triangle.

- START

- Input the the the length of triangle
 $a = 10$

- Calculate using the formula

$$A = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} 10^2$$

$$= 43.3$$

- Display the output

Area of an equilateral triangle is 43.3

11) Find the area of Circumference of circle

- START

- Input the radius
 $r = 5$

- Compute the circumference using the formula

$$C = 2\pi r$$

$$= 2 \times 3.14 \times 5$$

$$= 31.4$$

- Display the result
Circumference of the circle is 31.4

- STOP

12) Find the Volume of Sphere

- START

- Input the radius
 $r = 5$

- Compute by using the formula of volume of Sphere

$$V = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times 3.14 \times 5^3$$

$$= \frac{4}{3} \times 392.5$$

$$= 523.6$$

- Display the result
The volume of Sphere is 523.6

- STOP